Hyperspectral Target Detection Based on Generative Self-Supervised Learning With Wavelet Transform

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***Abstract*— Recently, generative self-supervised learning (GSSL) has gained extensive attention in hyperspectral remote sensing. For the hyperspectral target detection (HTD) task, traditional GSSL-based algorithms usually require hyperspectral images (HSIs) as additional datasets for pretraining, which are relatively resource-intensive and time-consuming. To better interpret the spectral–spatial information of HSIs while alleviating the dependence on large-scale hyperspectral datasets, we develop a novel two-stage framework for HTD based on GSSL in this article. In the preprocessing for the input HSI, a dimensional transformation (DT) module and a coarse detection reference (CDR) module are constructed to produce feature patches as training samples for subsequent pretraining and fine-tuning. In the pretraining stage for spectral–spatial reconstruction, we construct an asymmetric autoencoder (AE) architecture which leverages the transformer blocks with long-range perception to extract generalized features and explore discriminative feature representations of the input HSI. Specifically, a dual-stream wavelet patch embedding (DWPE) module is proposed to integrate the wavelet transform (WT) mechanism with the convolutional neural networks (CNNs), which extracts robust spectral–spatial features by performing convolutional operations with different frequency components of WT. In the fine-tuning stage, a novel signature-constrained cross-entropy (SC-CE) loss function is proposed to constrain the network optimization. For the final detection, a pixel-level fusion based on coarse detection based pixel-level fusion (CDPF) module is employed after inference to further suppress the interference from background. Experimental results on six real HSIs demonstrate that the proposed method achieves superior detection performance while maintaining the generalization of the pretrained model.**

***Index Terms*— Generative self-supervised learning (GSSL), hyperspectral image (HIS), hyperspectral target detection (HTD), remote sensing, wavelet transform (WT).**

I. Introduction

HYPERSPECTRAL remote sensing can acquire and record the spectral information within each pixel area across vast spectral wavelength bands, providing hyperspectral images (HSIs) that contain a large number of contiguous spectral bands [1]. Due to the high-spectral resolution, HSIs with the material discriminative ability are suitable for many applications, including image classification (IC) [2], target detection (TD) [3], and super resolution (SR) [4]. Hyperspectral TD (HTD) utilizes the spectral information of each pixel in a testing HSI to determine whether the pixel belongs to a certain material [5]. Specifically, given the reference spectrum of the target (i.e., target signature), HTD aims at finding and localizing targets in a specific HSI. Note that the reference spectra are usually obtained from a spectral library or from target pixels already identified in the scene [5].

In the last few decades, numerous classical methods have been developed for HTD. Initially, HTD was achieved using simple spectral matching methods based on spectral similarity. The spectral angle mapper (SAM) [6] and spectral information divergence (SID) [7] measure the distance between the testing pixel and the target signature to obtain matching scores. In addition, the structured background methods, such as the orthogonal subspace projection (OSP) [8] model, assume that any mixed pixel can be regarded as a linear combination for several endmembers with different abundances.

On this basis, many hypothesis testing methods based on statistical information and certain assumptions [9] have been proposed. For instance, the generalized likelihood ratio test (GLRT) [10] and its variants, such as the adaptive coherence estimator (ACE) [11] and the adaptive matched filter (MF) [12], have been proposed. Furthermore, the constrained energy minimization (CEM) [13] and its variants, such as hCEM [14] and E-CEM [15], are designed to minimize the response of backgrounds and maximize the response of targets.

Subsequently, several innovative techniques have been integrated into the classical methods. For example, the kernel-based methods, such as KCEM [16] and KOSP [17], map data to a high-dimensional space to achieve nonlinear transformation of the original features. Furthermore, the sparsity assumptions have been introduced to HTD to better utilize spectra. These models obtain detection results by constructing an appropriate dictionary and calculating the residuals between reconstructed pixels and the original pixels. Examples include the sparsity-based target detector (STD) [18] and the combined sparse and collaborative representation (CSCR) detector [19].

The above traditional methods often fail to extract the discriminative features in HSIs and rely on certain assumptions that may not match the practical application scenarios.

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Unlike traditional detectors, HTD models based on deep learning (DL) have gained attention and demonstrated effectiveness due to the impressive representation ability of neural networks, such as the autoencoder (AE), the convolutional neural network (CNN) [20], the generative adversarial network (GAN) [21], and the transformer [22]. However, the lack of labeled training samples is a major challenge for HTD since there are only one or a few priori target spectra available [23]. Therefore, many methods have been proposed to improve the training of DL-based HTD models, which adopt various sample expansion strategies to generate pseudolabeled samples.

In recent years, many supervised methods with training sample generation or augmentation based on the target signature have been developed. For instance, HTD-Net [24] extends the CNN model through an improved AE and linear prediction strategy, utilizing amounts of pseudolabeled samples for training. TSCNTD [25] employs sparse representation and a classification-based pixel selection strategy to generate a large number of pseudolabeled samples. Background learning based on target suppression constraint (BLTSC) [26] is trained on the pseudolabeled background samples to accurately learn the background distribution. Based on the transformer mechanism, both the HTD-TS^3 [27] and the HTDFormer [28] use generated pseudolabeled samples through a coarse detection and selection strategy to better extract spectral–spatial feature.

As for the unsupervised methods, the spectral regularized unsupervised network (SRUN) [29] is proposed to extract latent information from HSIs. The deep spatial–spectral network (DSN) [30] is designed by stacking multiple region-of-interest feature transformation (RFT)-multihead self-attention (MSA) (RM) blocks and deconvolution (DC) blocks. The interpretable representation network for HTD (HTD-IRN) [31] is constructed by a lightweight multiscale transformer, which can achieve adaptive learning of the background subspace.

Although the above DL-based methods obtain better detection performance than traditional methods, there still exists some limitations in these methods. The first limitation is the vulnerability of supervised detection to training samples. It is still challenging to produce abundant and accurate labeled samples for training in practice. The second limitation is that unsupervised methods usually perform TD only with the simple matching strategy. In addition, these DL-based methods struggle to fully extract spectral–spatial features in HSIs. Models based on CNN or the fully convolutional network (FCN) are severely constrained by high-computational complexity and sensor-specific transferability [27].

Nowadays, self-supervised learning (SSL) is emerging as a potential solution to the above limitations. Specifically, the SSL-based framework can capture the feature representations of the data without relying on labels, which provides an innovative approach to cope with the lacking of scarce labeled samples for supervised methods [23]. By training the model on the constructed valid posterior information, a valuable representation model can be obtained, which can get rid of the dependence of deep models on the prior information [32].

In summary, the primary goal of SSL is to generate robust representations independent of labeled samples. As the most representative SSL manner, the generative SSL (GSSL) can extract meaningful features by recovering the original data distribution without assumptions [33]. Currently, GSSL has become a key technique of foundation model (FM), achieving significant success in natural language processing (NLP) and computer vision (CV) [34]. The FM framework usually consists of a pretraining and a fine-tuning process, in which the pretraining is conducted via training model on massive unlabeled samples. In addition, the FM framework can leverage the learned universal knowledge across different tasks, which can significantly improve the cross-scenario generalization and migration capabilities. The architecture of FM usually adopts the transformer [22] mechanism as the backbone because of its scalability and flexibility, allowing for parameter scaling via the stacking of transformer blocks.

Recently, inspired by FM, researchers have explored the application of GSSL in remote sensing. The SatMAE [35] is a pretraining model for multispectral satellite images based on the masked AE (MAE) [36]. It adopts a masked image modeling (MIM) framework and predicts the content of masked regions based on information from visible regions. Although SatMAE can group adjacent spectral bands of multispectral data, it disrupts the continuity of the spectral dimension. The SpectralGPT [37] framework is pretrained on multispectral datasets, which divides the multispectral image into 3-D cubes and creates a visual embedding for MAE. However, these models are not suitable for processing serial hyperspectral sequences from HSI. Although Chen et al. [38] have proposed a pretraining model based on GSSL with spectral–spatial masking, it is pretrained on a dataset constructed for hyperspectral anomaly detection and evaluated on only four HSIs. In addition, existing GSSL-based HTD methods require large-scale HSI datasets for pretraining, which is relatively resource-intensive and time-consuming.

Therefore, to liberate the HTD model from dependence on the labeled training samples and enhance its feature extraction (FE) capability, we intend to design a novel two-stage HTD framework based on GSSL, which can realize effective FE and TD without additional large-scale HSI datasets. Specifically, the proposed framework needs to be implemented from the following two perspectives.

1) *Making Full Use of Input HSI:* Unlabeled samples used for GSSL can be generated within the input HSI of a specific HTD task after the appropriate preprocessing.
2) *Defining Tasks of Two Stages:* The task of the pretraining and fine-tuning should be clear. For the former, we are inspired by the wavelet transform (WT), a powerful tool for signal processing and analysis, which can extract robust spectral–spatial features at different scales and orientations [39]. For the latter, we tend to constrain the network optimization by designing a novel loss function.

As for the WT mechanism, it is a potential solution to extract spectral–spatial feature as a preprocessing operation. Recently, some researchers have exploited the properties of WT and combined WT with the CNNs in some tasks in hyperspectral remote sensing. Yang et al. [40] predict the wavelet coefficients of HSI with high resolution for hyperspectral super-resolution (HSR) task. Anand et al. [41] employ the discrete wavelet transform (DWT) as a preprocessing operation

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and then send extracted features to a specific classifier. Xu et al. [42] and Cao et al. [43] perform the 3-D DWT and the 3-D CNN separately for hyperspectral IC (HIC), where 3-D DWT is a preprocessing operation. Ahmad et al. [44] propose the WaveFormer to unify the downsampling with WT to decompress feature maps without loss. These works illustrate the benefits of integrating the properties of WT and CNNs in a single architecture for extracting robust spectral–spatial features.

Although some methods have introduced the WT mechanism into SR [40] task and the IC [41], [42], [43], [44] task of HSIs, the frameworks in these methods are not suitable for HTD tasks due to the differences in prior knowledge, inference objective, and training strategy. It is crucial to extract the spectral–spatial features in HTD tasks, which can determine the quality of the detection performance. Therefore, for the first time, we introduce the WT mechanism into HTD tasks in this article to extract spectral–spatial features through GSSL. In addition, considering that in the existing GSSL-based two-stage methods [37], [38], the pretraining stage for hyperspectral FE is relatively time-consuming and resource-intensive due to the requirement for large-scale datasets, which may be redundant and not necessary for a single HTD task. Hence, we propose a novel HTD framework based on GSSL with WT mechanism to fully exploit the input HSI through constructing adequate training samples for both pretraing and fine-tuning, achieving improved detection performance. The contributions of this article can be summarized as follows.

1) We propose a novel two-stage framework for HTD based on GSSL to focus on spectral–spatial FE. In the preprocessing, a dimensional transformation (DT) module and a coarse detection reference (CDR) module are designed to produce feature patches as training samples for subsequent pretraining and fine-tuning by only using the input HSI, alleviating the dependence on additional large-scale datasets for pretraining.
2) During pretraining for spectral–spatial reconstruction, we develop an asymmetric AE architecture with stacked transformer blocks to explore discriminative feature representations of the input HSI. Specifically, a dual-stream wavelet patch embedding (DWPE) module is proposed to integrate CNNs with the WT mechanism to extract robust spectral–spatial features, which perform convolutional operation with different frequency components of WT.
3) In the fine-tuning and detection stage, a novel signature-constrained cross-entropy (SC-CE) loss function is designed to constrain the network optimization. And a pixel-level fusion based on coarse detection based pixel-level fusion (CDPF) module is employed to suppress the interference of the background.

The remainder of this article is structured as follows. Section II provides a detailed description of the proposed method. The experimental results and corresponding analysis are presented in Section III. Finally, the conclusion is drawn in Section IV.

II. Method

This section demonstrates the proposed GSSL-based HTD framework, which consists of three main components: 1) the preprocessing of the input HSI; 2) the pretraining stage based on GSSL; and 3) the fine-tuning and detection stage. Fig. 1 shows the overall architecture of the proposed framework.

A. Preprocessing

For pretraining, existing GSSL-based methods usually utilize large-scale datasets containing numerous unlabeled samples, which may not be available or constructed easily. Therefore, to alleviate the dependence on additional datasets for training, we design a novel preprocessing strategy at first, which consists of the DT module and the CDR module.

In preprocessing, as shown in Fig. 1(a), the input HSI will be transformed into two groups of feature patches with fixed patch size and overlapping regions, which provides training samples for pretraining and fine-tuning, respectively.

1) DT Module: The DT module is employed to produce a padded feature tensor based on the input HSI. The padded feature tensor can be transformed into two groups of feature patches, which can be used as training samples for pretraining and fine-tuning. Specifically, the DT module includes three steps: principal component analysis (PCA), normalization, and mirror padding.

Considering different HSIs may contain hundreds of bands with different spectral resolutions and ranges, the pretraining on numerous HSIs is always challenging. Therefore, the dimensionality reduction (DR) is commonly performed on HSI to facilitate the reduction of noise and redundant information and improve the validity of distance or similarity [45].

In practice, the DR operation for HSI is usually realized via the band selection (BS) or the FE. The BS only selects a subset of bands from a full set of bands, which may result in losing important spectral information. Differently, FE aims to find a mapping from the original high-dimensional feature space to a lower dimensional space, which has a strong ability to extract spectral features while retaining spectral information. Therefore, PCA [46], one of the typical FE algorithms, is employed as the first operation in the DT module.

Specifically, given an input HSI $\mathbf{X}_C \in \mathbb{R}^{H\times W\times C}$, the PCA processes each spectrum of $\mathbf{X}_C$ and finally generates the output feature tensor $\mathbf{X} \in \mathbb{R}^{H\times W\times L}$.

The process can be described as follows:

$$\mathbf{X} = \mathrm{PCA}(\mathbf{X}_C) \tag{1}$$

where L, i.e., the number of principal components, is the length of each feature vector in $\mathbf{X}$ after mapping via PCA. Also, the number of feature vectors in $\mathbf{X}$ is $H \times W$.

Note that the number of principal components L is a critical parameter and comprehensive experiments with analysis in Section III are employed to identify the optimal L.

The second operation in the DT module is the normalization of the feature tensor $\mathbf{X}$. As channelwise normalization offers a better pixel value distribution compared with the normalization across all channels [47], we apply it to each channel of $\mathbf{X}$, yielding the normalized result $\mathbf{X}^n \in \mathbb{R}^{H\times W\times L}$.

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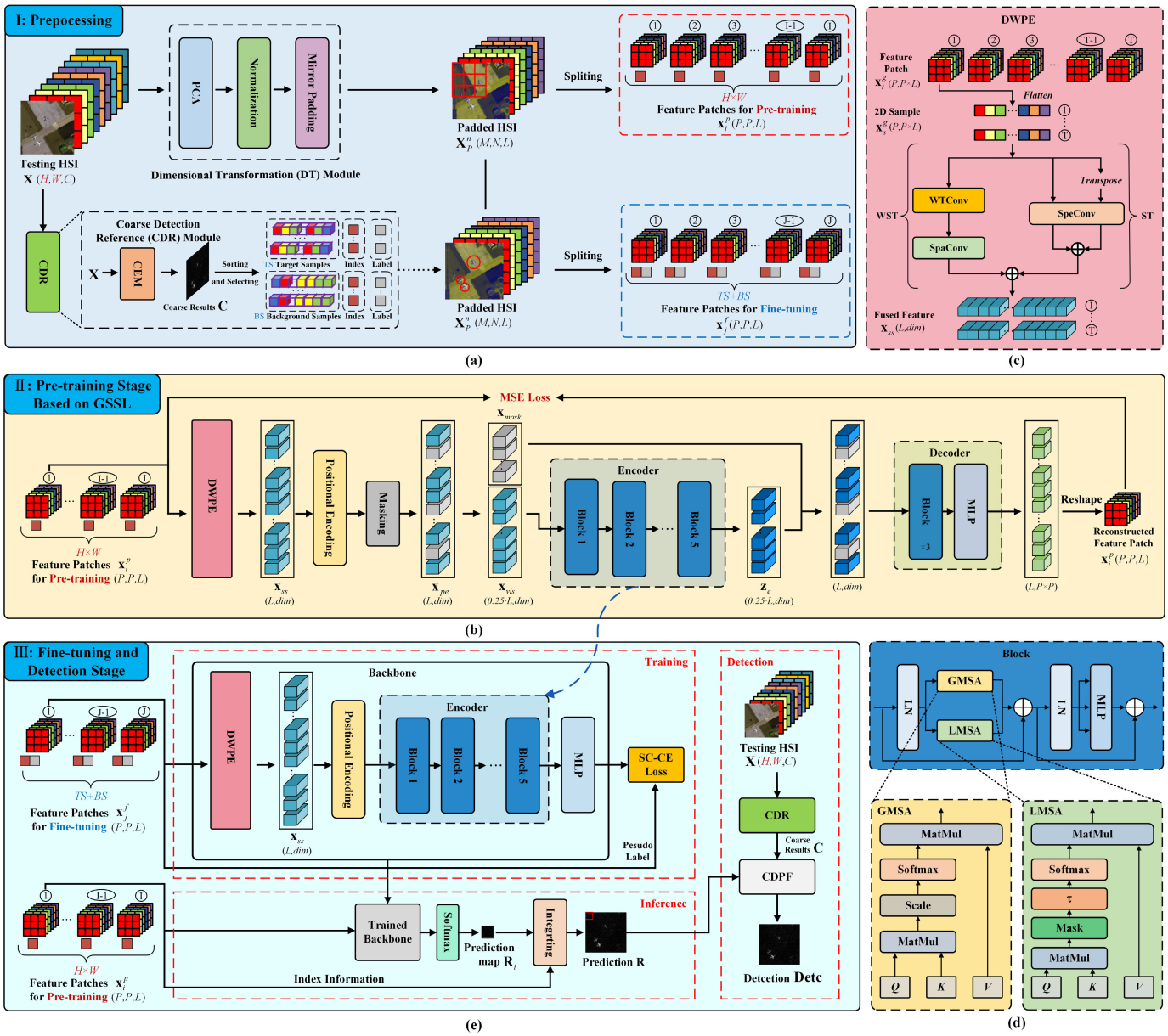


Fig. 1. Overall architecture of the proposed framework. (a) *In the preprocessing*, the input HSI is split into two groups of the feature patches, which serve as training samples for subsequent pretraining and fine-tuning. (b) *In the pretraining stage based on GSSL*, an asymmetric AE is used to reconstruct the input feature patch, which follows the principle of MIM. (c) *In the DWPE module*, each feature patch is transformed as the fused feature $\mathbf{x}_{ss}$ containing the spectral–spatial information extracted by the WST branch and the ST branch. (d) *Transformer block in AE*, which uses a GLMSA mechanism. (e) *In the fine-tuning and detection stage*, the pretrained encoder is transferred to the HTD task on the input HSI, which includes three steps based on coarse detection: training, inference, and detection.

The channelwise normalization can be formulated as follows:

$$\mathbf{X}_l^n = \text{Norm}(\mathbf{X}_l) = \frac{\mathbf{X}_l - \mathbf{X}_l^{\min}}{\mathbf{X}_l^{\max} - \mathbf{X}_l^{\min}}, \quad l = 1, 2, \ldots, L \tag{2}$$

where $\mathbf{X}_l$ and $\mathbf{X}_l^n$ represent the image matrices of $\mathbf{X}$ and $\mathbf{X}^n$ at the lth channel, respectively. $\mathbf{X}_l^{\max}$ and $\mathbf{X}_l^{\min}$ are the maximum and minimum values of image matrix $\mathbf{X}_l$.

The final step in the DT module is the mirror padding operation for the normalized feature tensor $\mathbf{X}^n$, which produces the padded feature tensor $\mathbf{X}_p^n$ for subsequent tensor splitting. Specifically, as shown in Fig. 2, the edges of the image are extended by mirror-reflecting the pixels at the original edge region, which is different from the widely used zero padding. After the mirror padding, a boundary area around the original image will be added. Also, the width of the added boundary region is set as $(P/2)$, which is a critical parameter that can be affected by the feature patch size P. Experiments with analysis in Section III are employed to identify the optimal patch size.

As shown in Fig. 2, by making the padded pixels similar to the original pixels, the mirror padding operation can maintain the context of pixels at the edge region of the original image.

The mirror padding process for $\mathbf{X}^n$ can be described as follows:

$$\mathbf{X}_p^n = \text{Padding}\left(\mathbf{X}^n\right) \tag{3}$$

where $\mathbf{X}_p^n \in \mathbb{R}^{M \times N \times L}$, $M = H + P$ and $N = W + P$.

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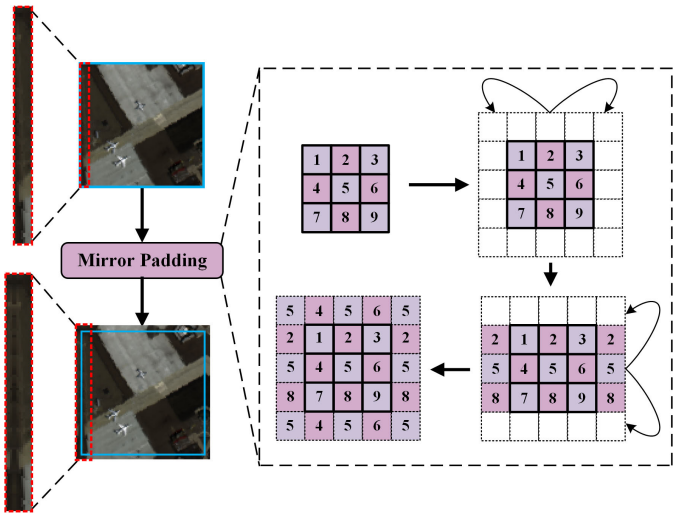


Fig. 2. Diagram of the mirror padding operation. For a padded image matrix, the top and bottom rows are created by mirroring the second and second-last rows of the original image. Also, the left and right columns are created by mirroring the second and second-last columns of the original image.

Then, the padded feature tensor $\mathbf{X}_p^n$ will be split into feature patches to make up two groups of feature patches, which can be used as training samples for pretraining and fine-tuning, respectively. Note that the patches in each group have a fixed patch size and overlapping region.

Specifically, in the padded feature tensor $\mathbf{X}_p^n$, each feature vector not on the added boundary region will be selected as a central vector at first. Then, a square window with size P is selected around the central vector. Finally, all feature vectors located in the window are selected as a square feature patch. Therefore, a group of feature patches can be obtained as follows: $P_p = \{\mathbf{x}_1^p, \mathbf{x}_2^p, \ldots, \mathbf{x}_i^p, \ldots, \mathbf{x}_I^p\}$, where $\mathbf{x}_i^p \in \mathbb{R}^{P\times P\times L}$ and $I = H \times W$. At the same time, the index information of each feature patch has been recorded. These feature patches will be used as the training samples for pretraining.

As for the training samples for fine-tuning, only feature vectors with pseudolabels are selected as central vectors to produce feature patches. Also, the pseudolabels can be derived from the CDR module.

2) CDR Module: Since HSI contains sparse target pixels and abundant background pixels, imbalanced category may greatly affect detection performance. Therefore, the CDR module is adopted to provide adequate training samples with pseudolabels for fine-tuning.

Specifically, the CDR module utilizes the classic detector CEM [13] to perform coarse detection on the input HSI $\mathbf{X}_C$. Constrained by the target signature $\mathbf{d}$, the goal of CEM is to design a finite impulse response (FIR) linear filter $\mathbf{w}$ that ensures the output of $\mathbf{d}$ as a constant, i.e., $\mathbf{d}^{\mathrm{T}}\mathbf{w} = 1$, while minimizing the average output energy of the filter, i.e., $E = \mathbf{w}^{\mathrm{T}}\mathbf{R}\mathbf{w}$, where $\mathbf{R}$ is the data sample correlation matrix.

The optimal solution can be derived as follows:

$$\mathbf{w}^{\mathrm{opt}} = \frac{\mathbf{R}^{-1}\mathbf{d}}{\mathbf{d}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{d}}. \tag{4}$$

Based on the results of coarse detection, the top TS pixels with high probabilities are set as target samples, while the bottom BS pixels with low probabilities are set as background samples. In this way, TS+BS pseudosamples with pseudolabels can be obtained with label and index information.

In the CDR module, the number of pseudotarget samples TS is set to 1.5% of the number of pixels in the input HSI. And the number of pseudobackground samples BS is set to 50% of the number of pixels in the input HSI. Note that the selection ratio directly affects whether the target and background samples selected by CEM are correct. Also, the ratio for selecting pseudobackground samples and pseudotarget samples is usually empirical and not fixed. A range of the most appropriate ratios exists for each dataset. Also, the above ratio is relatively suitable for specific datasets in this article.

According to the label information of pseudosamples, the corresponding feature vectors in the padded feature tensor $\mathbf{X}_p^n$ can be labeled as target or background vectors. Each labeled feature vector will be processed to produce a square feature patch. In this way, a group of feature patches can be obtained, which is denoted by $P_f = \{\mathbf{x}_1^f, \mathbf{x}_2^f, \ldots, \mathbf{x}_j^f, \ldots, \mathbf{x}_J^f\}$ where $\mathbf{x}_j^f \in \mathbb{R}^{P\times P\times L}$ and $J = \mathrm{TS}+\mathrm{BS}$. These feature patches will be used as training samples for fine-tuning. The label and index information of each feature patch have been recorded.

Note that the same targets in an HSI may have different spectra because of the spatial and temporal variation, especially when the number of bands is high. In this case, the CEM detector can only guarantee the output of signature $\mathbf{d}$ constant, regardless of other targets with the spectral signature similar to $\mathbf{d}$, which implies not all targets can be detected in coarse detection. It means that training samples constructed based on coarse detection may be incompletely and inaccurately labeled. Therefore, the fine-tuning and detection stage can also be considered as a process under inaccurate supervision, one of the three types of weak supervision [48].

B. Pretraining Stage Based on GSSL

To realize GSSL based on the feature patches from preprocessing, an asymmetric AE is designed for spectral–spatial reconstruction of feature patches, which follows the principle of MIM proposed in MAE [36].

As shown in Fig. 1(b), the pretraining stage consists of four components: the DWPE module, the masking strategy, the AE network, and the reconstruction loss.

1) Dual-Stream Wavelet Patch Embedding: To exploit comprehensive spectral–spatial representation, the DWPE module is proposed, as shown in Fig. 1(c). The DWPE module aims to transform feature patches into fused features to represent the spectral–spatial information via the designed wavelet-spatial transform (WST) branch and spectral transform (ST) branch.

Given two groups of feature patches P_p and P_f, which are utilized for pretraining and fine-tuning, respectively, let $\mathbf{x}_t^g \in \mathbb{R}^{P\times P\times L}$ be a feature patch, where g represents the corresponding group. The feature patch $\mathbf{x}_t^g$ with L channels can be regarded as L image matrices. Each image matrix can be denoted by $\mathbf{x} \in \mathbb{R}^{P\times P}$. To extract the spectral–spatial information of $\mathbf{x}_t^g$, L image matrices of $\mathbf{x}_t^g$ are concatenated to produce a 2-D sample denoted by $\mathbf{x}_s^g \in \mathbb{R}^{P\times(P\times L)}$. Then, each

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2-D sample will be fed into the WST branch and the STB branch at the same time.

a) WST branch: To extract spatial features of the input 2-D sample, we construct the WST branch including the wavelet transform convolution (WTConv) layer and the spatial convolution (SpaConv) layer.

As shown in Fig. 1(c), the WTConv layer first decomposes the input 2-D sample into different frequency maps using WT. Then, the convolutional operation with small kernels is applied to both low-frequency and high-frequency maps to cover a wide receptive field, ensuring robust responses to all frequency information. Also, the reconstructed output is subsequently obtained using the inverse wavelet transform (IWT). This reconstructed output is then added to the directly convolved 2-D sample to produce the enhanced output, which will be processed by the SpaConv layer finally to improve the spatial feature representation.

Specifically, given a 2-D sample $\mathbf{x}_s^g$, we employ the Haar WT because it is efficient and straightforward [49], [50]. For 2-D one-level Haar WT, the operation is applied to height and width dimensions, resulting in a convolution with a stride of 2 using the filter set f composed of following four filters:

$$f_{LL} = \frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}, \quad f_{LH} = \frac{1}{2}\begin{bmatrix}1 & -1\\ 1 & -1\end{bmatrix}$$
$$f_{HL} = \frac{1}{2}\begin{bmatrix}1 & 1\\ -1 & -1\end{bmatrix}, \quad f_{HH} = \frac{1}{2}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix} \tag{5}$$

where f_{LL} is a low-pass filter and f_{LH}, f_{HL}, and f_{HH} form a set of high-pass filters.

Then, the output of the convolution can be expressed as follows:

$$\left[\mathbf{x}_{LL}^g, \mathbf{x}_{LH}^g, \mathbf{x}_{HL}^g, \mathbf{x}_{HH}^g\right] = \mathrm{Conv}\left(f, \mathbf{x}_s^g\right) = \mathrm{WT}\left(\mathbf{x}_s^g\right) \tag{6}$$

where $\mathbf{x}_{LL}^g$ represents the low-frequency map of $\mathbf{x}_s^g$ and $\mathbf{x}_{LH}^g$, $\mathbf{x}_{HL}^g$, and $\mathbf{x}_{HH}^g$ are its horizontal, vertical, and diagonal high-frequency maps. Note that each frequency map has half the spatial resolution of $\mathbf{x}_s^g$.

Then, a small-kernel depthwise convolution with a stride of 1 is performed on different frequency maps. Also, IWT is used to reconstruct the output, which can be described as follows:

$$\mathbf{x}_r^g = \mathrm{IWT}\left(\mathrm{Conv}\left(W, \mathrm{WT}\left(\mathbf{x}_s^g\right)\right)\right) \tag{7}$$

where W is the weight tensor of four 3×3 depthwise kernels, which has the same number as the input channels of $\mathbf{x}_t^g$.

This process not only isolates the convolution for each frequency map but also allows a small kernel to cover a large region of the original input, expanding its receptive field.

Finally, the enhanced output is constructed by adding $\mathbf{x}_r^g$ with the directly convolved 2-D sample as the following process:

$$\mathbf{x}_e^g = \mathbf{x}_r^g + \mathrm{Conv}\left(W, \mathbf{x}_s^g\right). \tag{8}$$

Next, the enhanced output $\mathbf{x}_e^g$ is fed into the SpaConv layer, which is designed to further obtain the spatial feature $\mathbf{x}_{\mathrm{spa}}^g$. The process can be described as follows:

$$\mathbf{x}_{\mathrm{spa}}^g = \mathrm{SpaConv}\left(\mathbf{x}_e^g\right) \tag{9}$$

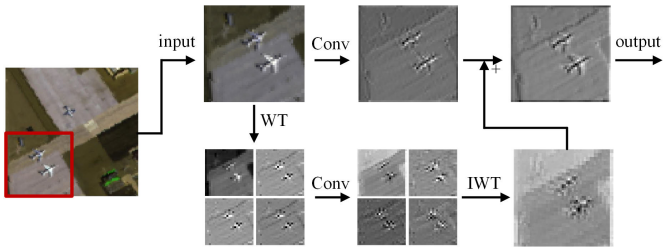


Fig. 3. Diagram of the local transformation of the 2-D one-level Haar WT on an entire image.

where $\mathbf{x}_{\mathrm{spa}}^g \in \mathbb{R}^{L\times \mathrm{dim}}$ because the kernel size and stride of the convolutional operation in SpaConv layer are both (P, P), and the number of convolutional kernels is dim.

Fig. 3 visualizes the local transformation of the 2-D one-level Haar WT on an entire image. Note that Haar WT exhibits locality and composability. We apply Haar WT to the entire 2-D sample rather than performing multiple local transformations on the 2-D sample. Although both approaches are equivalent, the former is generally more efficient, as it avoids multiple repeating computations and memory overflow.

b) ST branch: The ST branch is designed to extract spectral features of the input 2-D sample, which is realized by a spectral convolution (SpeConv) layer.

Given a 2-D sample $\mathbf{x}_s^g$, we transpose it to obtain the transposed input $\mathbf{x}_s^{g\mathrm{T}}$ at first. Then, the original input and transposed input are fed into the SpeConv layer together, and their outputs are added to generate the spectral feature $\mathbf{x}_{\mathrm{spe}}^g$ as follows:

$$\mathbf{x}_{\mathrm{spe}}^g = \mathrm{SpeConv}\left(\mathbf{x}_s^g\right) + \mathrm{SpeConv}\left(\mathbf{x}_s^{g\mathrm{T}}\right) \tag{10}$$

where $\mathbf{x}_{\mathrm{spe}}^g \in \mathbb{R}^{P\times((L/P)\times \mathrm{dim})}$ because the kernel size and stride of the convolution in SpeConv layer are both $(1, P^2)$, and the number of convolutional kernels is dim.

Finally, the spatial feature $\mathbf{x}_{\mathrm{spa}}^g$ and the spectral feature $\mathbf{x}_{\mathrm{spe}}^g$ are concatenated to produce the fused feature $\mathbf{x}_{\mathrm{ss}} \in \mathbb{R}^{L\times \mathrm{dim}}$, which contains L spectral–spatial vectors.

The process can be expressed as follows:

$$\mathbf{x}_{\mathrm{ss}} = \mathrm{GELU}\left(\mathrm{LN}\left(\mathbf{x}_{\mathrm{spa}}^g + \mathbf{x}_{\mathrm{spe}}^g\right)\right). \tag{11}$$

2) Masking Strategy: Before the masking operation, the fused feature $\mathbf{x}_{\mathrm{ss}}$ is added to a learnable positional encoding weights $\mathbf{E}_{\mathrm{pos}}$, which can be expressed as follows:

$$\mathbf{x}_{\mathrm{pe}} = \mathrm{PE}(\mathbf{x}_{\mathrm{ss}}) = \mathbf{x}_{\mathrm{ss}} + \mathbf{E}_{\mathrm{pos}} \tag{12}$$

where $\mathbf{E}_{\mathrm{pos}} \in \mathbb{R}^{L\times \mathrm{dim}}$ and $\mathbf{x}_{\mathrm{pe}} \in \mathbb{R}^{L\times \mathrm{dim}}$.

Then, the spectral–spatial vectors in $\mathbf{x}_{\mathrm{pe}}$ are randomly masked with a mask ratio Q, where the values of masked vectors are replaced by zeros. Note that Q is a critical parameter and comprehensive experiments with analysis in Section III are employed to identify the optimal Q.

After the masking operation, the spectral–spatial vectors in $\mathbf{x}_{\mathrm{pe}}$ are divided into the visible and masked vectors denoted by T_{vis} and T_{mask}, respectively. In practice, the vectors in T_{vis} and T_{mask} are concatenated for convenience, respectively. Note that the mapped dimensions (i.e., dim) of each vector will not change during this process.

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The above processes can be expressed as follows:

$$\{T_{\text{vis}}, T_{\text{mask}}\} = \text{Mask}(\mathbf{x}_{\text{pe}}) \tag{13}$$

$$\mathbf{x}_{\text{vis}} = \text{Concat}(T_{\text{vis}}) \tag{14}$$

$$\mathbf{x}_{\text{mask}} = \text{Concat}(T_{\text{mask}}) \tag{15}$$

where $\mathbf{x}_{\text{vis}} \in \mathbb{R}^{Q_1 \times \text{dim}}$ and $\mathbf{x}_{\text{mask}} \in \mathbb{R}^{Q_2 \times \text{dim}}$ represent the visible fused feature and the masked fused feature, respectively. Also, Q_1 and Q_2 are the number of visible and masked spectral–spatial vectors in $\mathbf{x}_{\text{pe}}$, i.e., $Q = Q_1 + Q_2$.

The masking operation can enhance the sensitivity of the pretraining model to input variations, enabling it better adapt to changes in the input data [51].

3) AE Network: In this article, an asymmetric AE with stacked vision transformers (ViTs) is constructed to realize spectral–spatial reconstruction of the input feature patches.

Considering the traditional multihead self-attention (MSA) mechanism in ViTs faces challenges in capturing local detailed features and extracting texture patterns, we adopt the global and local MSA (GLMSA) mechanism [52] in this work.

The illustration of the transformer block in the proposed framework is shown in Fig. 1(d). A block consists of a GLMSA layer and a multilayer perceptron (MLP) layer.

a) Encoder: The encoder consists of five transformer blocks with four attention heads in MSA. Only the visible fused feature $\mathbf{x}_{\text{vis}}$ are processed through the encoder.

The processing of the encoder can be described as follows:

$$z_e = \text{Encoder}(\mathbf{x}_{\text{vis}}) \tag{16}$$

where z_e is encoded feature with the same dimension as $\mathbf{x}_{\text{vis}}$.

GLMSA consists of the global MSA (GMSA) and the local MSA (LMSA), which are beneficial for distinguishing and integrating local features and global features.

Given the visible fused feature $\mathbf{x}_{\text{vis}}$, using learnable projection matrices $\mathbf{W}_Q$, $\mathbf{W}_K$, and $\mathbf{W}_V$, $\mathbf{x}_{\text{vis}}$ can be linearly transformed to query ($\mathbf{Q}$), key ($\mathbf{K}$), and value ($\mathbf{V}$) matrices.

For GMSA, the output is computed as a weighted sum of values ($\mathbf{V}$), where the weight assigned to each value ($\mathbf{V}$) is determined by a softmax function with a dot product scaled by $(1/\sqrt{d_k})$ between the query ($\mathbf{Q}$) and key ($\mathbf{K}$).

The computing process can be formulated as follows:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{soft}\max\left(\frac{\mathbf{Q}\mathbf{K}^{\text{T}}}{\sqrt{d_k}}\right)\mathbf{V} \tag{17}$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ represent the query matrix, key matrix, and value matrix. Also, d_k is the dimension of the input.

GMSA with m heads can be formulated as follows:

$$\text{GMSA}(\mathbf{x}_{\text{vis}}) = \text{Concat}(\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_m)\mathbf{W}^O \tag{18}$$

where $\mathbf{h}_m = \text{Attention}(\mathbf{x}_{\text{vis}}\mathbf{W}_m^Q, \mathbf{x}_{\text{vis}}\mathbf{W}_m^K, \mathbf{x}_{\text{vis}}\mathbf{W}_m^V)$ and $\mathbf{W}^O$ represents the corresponding learned matrix.

For LMSA, the matrix $\mathbf{M}$ is obtained by a dot product between the linearly transformed matrices $\mathbf{Q}$ and $\mathbf{K}$. The position of the matrix $\mathbf{M}$ is defined as $\mathbf{M}_{u,v} \in \mathbb{R}^{L \times L}$, and the process is formulated as follows:

$$F_{\text{LSA}}(\mathbf{Q}, \mathbf{K}) = \begin{cases} \mathbf{M}_{u,v}(u \neq v) \\ -\infty(u = v). \end{cases} \tag{19}$$

To mitigate self-relationship, the diagonal elements of the matrix $\mathbf{M}$ are masked. After that, the softmax operation with a learnable factor τ is applied to normalize the features of $F_{\text{LSA}}(\mathbf{Q}, \mathbf{K})$, which ensures LMSA to independently adjust the softmax temperature throughout the learning process. The process ca be expressed as follows:

$$\text{Att}_L(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{F_{\text{LSA}}(\mathbf{Q}, \mathbf{K})}{\tau}\right)\mathbf{V}. \tag{20}$$

LMSA with m heads can be formulated as follows:

$$\text{LMSA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\mathbf{h}_1, \mathbf{h}_1, \ldots, \mathbf{h}_m)\mathbf{W}^O \tag{21}$$

where $\mathbf{h}_m = \text{Att}_L(\mathbf{x}_{\text{vis}}\mathbf{W}_m^Q, \mathbf{x}_{\text{vis}}\mathbf{W}_m^K, \mathbf{x}_{\text{vis}}\mathbf{W}_m^V)$ and $\mathbf{W}^O$ represents the corresponding learned matrix.

b) Decoder: The decoder only consists of three transformer blocks and one linear projection layer. Each block adopts MSA with three attention heads. The decoder is designed to reconstruct the original feature patch from the encoded feature z_e and the masked fused feature $\mathbf{x}_{\text{mask}}$. Note that the masked fused feature $\mathbf{x}_{\text{mask}}$ are added to the encoded feature z_e based on the original position. Then, the final layer of the decoder is a linear projection layer to produce an output with multiple channels, where the number of channels is equal to the number of spectral matrices in the feature patch. Finally, this output is reshaped to reconstruct the original feature patch.

The process can be expressed as follows:

$$\hat{\mathbf{x}}_i^p = \text{Decoder}(z_e, \mathbf{x}_{\text{mask}}) \tag{22}$$

where $\hat{\mathbf{x}}_i^p$ represents the reconstructed feature patches.

4) Reconstruction Loss: To evaluate the quality of spectral–spatial reconstruction, we compare the difference between the reconstructed and original feature patches. Specifically, the reconstruction loss is calculated using MSE.

The calculational process can be expressed as follows:

$$\mathcal{L}_{\text{mse}} = \frac{1}{I}\sum_{i=1}^{I}\left(\hat{\mathbf{x}}_i^p - \mathbf{x}_i^p\right)^2 \tag{23}$$

where $\hat{\mathbf{x}}_i^p$ is a reconstructed feature patch and $\mathbf{x}_i^p$ is the corresponding original feature patch in P_p.

C. Fine-Tuning and Detection Stage

After pretraining, the pretrained encoder has obtained the ability to extract features from the input HSI. In this stage, the pretrained encoder will be transferred to the pixel-level TD. As shown in Fig. 1(e), based on the coarse detection, the fine-tuning and detection stage includes three main steps: training, inference, and detection.

1) Training: Unlike pretraining, fine-tuning utilizes the feature patches in P_f with label information as training samples, which helps the model learn the characteristics of target and background samples. Note that the label information is derived from coarse detection via the CDR module in preprocessing.

The backbone for fine-tuning consists of four parts: the DWPE module, the positional encoding module, the pretrained encoder, and an MLP layer. Note that the fused feature from the DWPE module will be directly fed into the pretrained

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encoder without masking. The positional encoding module is designed to record the positional information of each spectral–spatial vector in the visible fused feature. The pretrained encoder serves as a feature extractor, which can be directly used for fine-tuning, as the encoder processes each spectral–spatial vector within the fused feature, rather than the entire fused feature, and each vector has the same dimension. Also, the MLP layer is adopted to predict whether the label of each feature patch belongs to the target category.

To enhance the distinction between target and background samples, the SC-CE loss function is proposed for fine-tuning, which mitigates the interference of potential background samples. Specifically, based on the index information of the training samples for fine-tuning, the original pixel $\mathbf{x}_j$ corresponding to each training sample $\mathbf{x}_j^f$ in the input HSI can be identified. Note that the index information is derived from the coarse detection in the CDR module during preprocessing. Then, a coefficient α is introduced to constrain the CE loss, which calculates the difference between the feature patch $\mathbf{x}_j$ and the limited prior information $\mathbf{d}$. It should be clarified that α is not calculated using all training samples, but rather for each training sample. The coefficient α can control the direction of network optimization during the fine-tuning stage, i.e., highlighting the target while suppressing the background. Specifically, if a training sample is closer to the target sample, the value of α will be larger, resulting in a larger CE loss for that sample, and the network will optimize toward the target. Conversely, if a training sample is closer to the background sample, the value of α will be smaller, leading to a smaller CE loss for that sample, which will slow down the speed of network optimization toward the background. In addition, to prevent α from being 0, a coefficient λ is introduced in the calculation of α, which helps avoid excessive differences between background samples and the target spectrum $\mathbf{d}$ (i.e., preventing alpha from becoming too small). Through the experiments, λ is set to 0.01, which can allow the network optimization to proceed normally.

The constraint coefficient α can be expressed as follows:

$$\alpha = \exp\left[-\lambda \cdot \left(\mathbf{x}_j - \mathbf{d}\right)^2\right] \tag{24}$$

where λ is an adjustable parameter.

Also, the SC-CE loss can be computed as follows:

$$\mathcal{L}_{\text{SC-CE}} = \alpha \cdot \mathcal{L}_{\text{CE}} \tag{25}$$

$$\mathcal{L}_{\text{CE}} = \frac{1}{J}\sum_{j=1}^{J} -\alpha \cdot \left[y_j \cdot \log\left(p_j^1\right) + \left(1 - y_j\right) \cdot \log\left(p_j^2\right)\right] \tag{26}$$

where J is the number of training samples in P_f for fine-tuning and p_j^1 and p_j^2 are the prediction possibility of sample $\mathbf{x}_j$ to be target and background. Also, y_j represents the pseudolabel of sample $\mathbf{x}_j$ with values 1 for target or 0 for background.

2) Inference: In this step, feature patches for pretraining are used to represent the input HSI. These feature patches are fed into the trained backbone, as shown in Fig. 1(e). Then, a softmax function is used to limit the output probability to the range of [0, 1], generating the prediction maps of statistics values. After all feature patches have been processed to the trained backbone, the prediction maps $\mathbf{R}_i$ are integrated back to produce the prediction result $\mathbf{R}$ with the same dimension as the input HSI, which is based on the index information of each feature patch. Note that the index information are derived from the coarse detection in the CDR module during preprocessing.

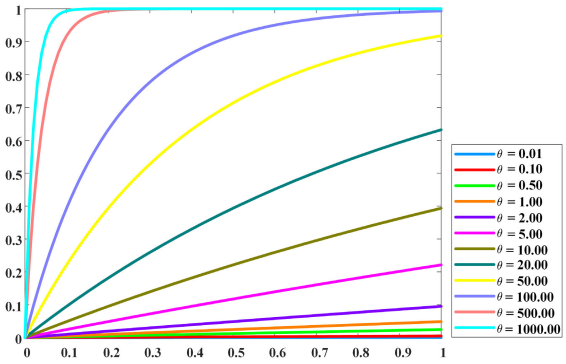


Fig. 4. Shape of exponential constrained nonlinear transform factor κ.

3) Detection: In this step, the pixel-level fusion based on CDPF module is designed to refine the prediction result via pixel-level fusion based on coarse detection. Specifically, the output $\mathbf{R}$ from inference is fed into the CDPF module together with the coarse detection result $\mathbf{C}$, which produces the final detection result **Detc**. Note that the coarse detection results $\mathbf{C}$ is derived from the CDR module during preprocessing. The CDPF module is based on an exponential-constrained nonlinear function, which can integrate features from the spectral domain and spatial domain while suppressing background interference.

This process can be expressed as follows:

$$\mathbf{Detc} = \kappa \odot \mathbf{R} = \left(\mathbf{1} - e^{-0.05\cdot\theta\cdot\mathbf{C}}\right) \odot \mathbf{R} \tag{27}$$

where θ is a positive parameter used to adjust the nonlinear transformation factor κ, which is aimed at using the coarse detection result $\mathbf{C}$ to improve detection performance.

Fig. 4 shows the shape of κ under different θ selections. It can be found that instances with a high probability of being targets are retained, and κ is relatively stable and close to 1. When κ is relatively stable and close to 1, instances with a high probability of being targets are retained. However, instances are suppressed when the probability of the background becomes high, and κ drops sharply and approaches 0.

III. Experiments

A. Datasets Description

As listed in Table I, the proposed method is evaluated on six real HSI datasets, which can be denoted by Airport [53], Beach [53], Urban [53], San Diego [15], HYDICE [53], and EI Segundo [54]. These datasets are captured at various scenes, including the airport area, the beach area, and the urban area. Also, these datasets are collected by different sensors, including the airborne visible/infrared imaging spectrometer (AVIRIS) sensor [55] and the hyperspectral

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TABLE I
INFORMATION OF HSI DATASETS

Dataset	Capture Place	Sensor	Background Scene	Target Type	Spatial Size	Target Size	Bands	Resolution
Airport	Los Angeles, USA	AVIRIS	airport area	airplanes	100×100 pixels	60 pixels	191	7.1m
Beach	Cat Island, Japan	AVIRIS	beach area	boats	150×150 pixels	19 pixels	188	17.2m
Urban	Texas coast, USA	AVIRIS	urban area	man-made objects	100×100 pixels	67 pixels	204	17.2m
San Diego	California, USA	AVIRIS	airport area	airplanes	100×100 pixels	134 pixels	189	3.5m
HYDICE	California, USA	HYDICE	urban area	cars and roofs	80×100 pixels	19 pixels	175	1m
EI Segundo	California, USA	AVIRIS	beach area	tanks and towers	250×300 pixels	2048 pixels	224	7.1m

digital imagery collection experiment (HYDICE) airborne sensor [56]. Multiscale targets of different categories and structures are distributed in these datasets, such as airplanes, boats, and man-made objects.

B. Experimental Settings

1) Implementation Details: For parameters proposed in our method, during the preprocessing, in the DT module, the length of each feature vector L is set to 20, and the feature patch size P is set to 9. In the CDR module, the number of pseudotarget samples TS is set to 1.5% of the number of pixels in the input HSI. And the number of pseudobackground samples BS is set to 50% of the number of pixels in the input HSI. In the pretraining stage, the mask ratio of spectral masking Q is set as 0.7, the batch size is set as 128, and the number of epochs is set as 200. In the fine-tuning and detection stage, the adjustable parameter λ in SC-BCE loss is set as 0.01, the batch size is set as 1024, and the number of epochs is set as 100. The optimizer and learning rate in the two stages are set as AdamW and $5e^{-4}$, respectively. All experiments were implemented on an Ubuntu 22.10 LTS operating system (OS) with hardware configuration of a six-core 12th Generation Intel[1] Core[2] i5-12400F CPU operating at 2.5 GHz and 32-GB RAM and two NVIDIA GeForce RTX 3090 GPUs with 24-GB memory. Compiling softwares consists of a Visual Studio Code 2024 version 1.95 and a MATLAB R2023a.

2) Comparative Methods: To evaluate the effectiveness of the proposed method, we compare it with eight different HTD methods. There are six classic traditional detection methods including SAM [6], ACE [11], CEM [13], hCEM [14], E-CEM [15], CSCR [19], and two DL-based methods containing BLTSC [26] and HTD-IRN [31]. These methods are popular as their outstanding performance. In addition, there are parameter settings of different HTD methods excluding SAM, ACE, and CEM, as these methods have no adjustable parameters. Then, for hCEM on the San Diego, HYDICE, and EI Segundo datasets, the parameters λ and ϵ are set to 100 and $1e^{-2}$, respectively. For hCEM on other datasets, the parameters λ and ϵ are set to 200 and $1e^{-6}$, respectively. For E-CEM, the number of detection layers, the number of CEM detectors per layer, and the regularization coefficient are set to 10, 6, and $1e^{-6}$, respectively. For CSCR, the parameters λ, σ, and β in CSCR are set as 0.1, 1.5, and 2.5, respectively. For BLTSC, the coarse detection is based on a CEM detector, and the learning rate and epoch are set to $1e^{-4}$ and 500, respectively.

[1]Registered trademark.
[2]Trademarked.

3) Evaluation Metrics: In order to evaluate the performance of the proposed method, we use the receiver operating characteristic (ROC) curve [57] and the area under the curve (AUC) [58] for quantitative analysis. ROC curves are extensively utilized as an evaluation method for the HTD task, which illustrates the dynamic relationship between the true positive rate (TPR) and false positive rate (FPR) based on the reference map. This relationship is visualized by plotting TPR against FPR at various thresholds τ. The AUC metric can be derived from the ROC curve by calculating the entire area under it, which provides a comprehensive measure of detection performance. Specifically, $\mathrm{AUC}_{(D,F)}$ and $\mathrm{AUC}_{(F,\tau)}$ are key metrics for evaluating the performance of a detector from aspects including the effectiveness, the detection probability, and background suppression (BS) capability of a detector. Generally, the higher $\mathrm{AUC}_{(D,F)}$ and the smaller $\mathrm{AUC}_{(F,\tau)}$ indicate superior detector performance. Note that we use AUC_1 and AUC_2 to represent $\mathrm{AUC}_{(D,F)}$ and $\mathrm{AUC}_{(F,\tau)}$, respectively. In addition, to evaluate the effectiveness of the proposed method in separating the target from the background, we use the target–background separability plots [54] to show the separation degree of the target and background. In these plots, the corresponding values of the target and background are extracted from the detection results. The red box and green box represent the target and background, respectively. The larger the distance between the red and green boxes, the higher the separation of the target and background.

C. Detection Performance

To evaluate the detection performance of the proposed method, eight detection methods are selected for comparison, including six traditional methods and two DL-based methods.

The visual detection results in Fig. 5 reveal that the detection results obtained by the proposed method are closest to the ground truth. Specifically, on six HSI datasets, hCEM produces detection results with much noise, which makes it difficult to distinguish between background and target pixels. CSCR can only detect partial targets that are not fully separated from the background. The classical methods SAM and CEM can successfully identify most target pixels at the cost of high FPRs, where a large number of background pixels are falsely detected as the targets. Although ACE, E-CEM, and BLTSC can effectively suppress the background, targets cannot be detected completely. HTD-IRN can detect targets clearly but struggles to suppress the backgrounds well. The proposed method can effectively highlight the targets with different sizes and shapes while suppressing the complex background. In particular, on the EI Segundo dataset, the proposed method

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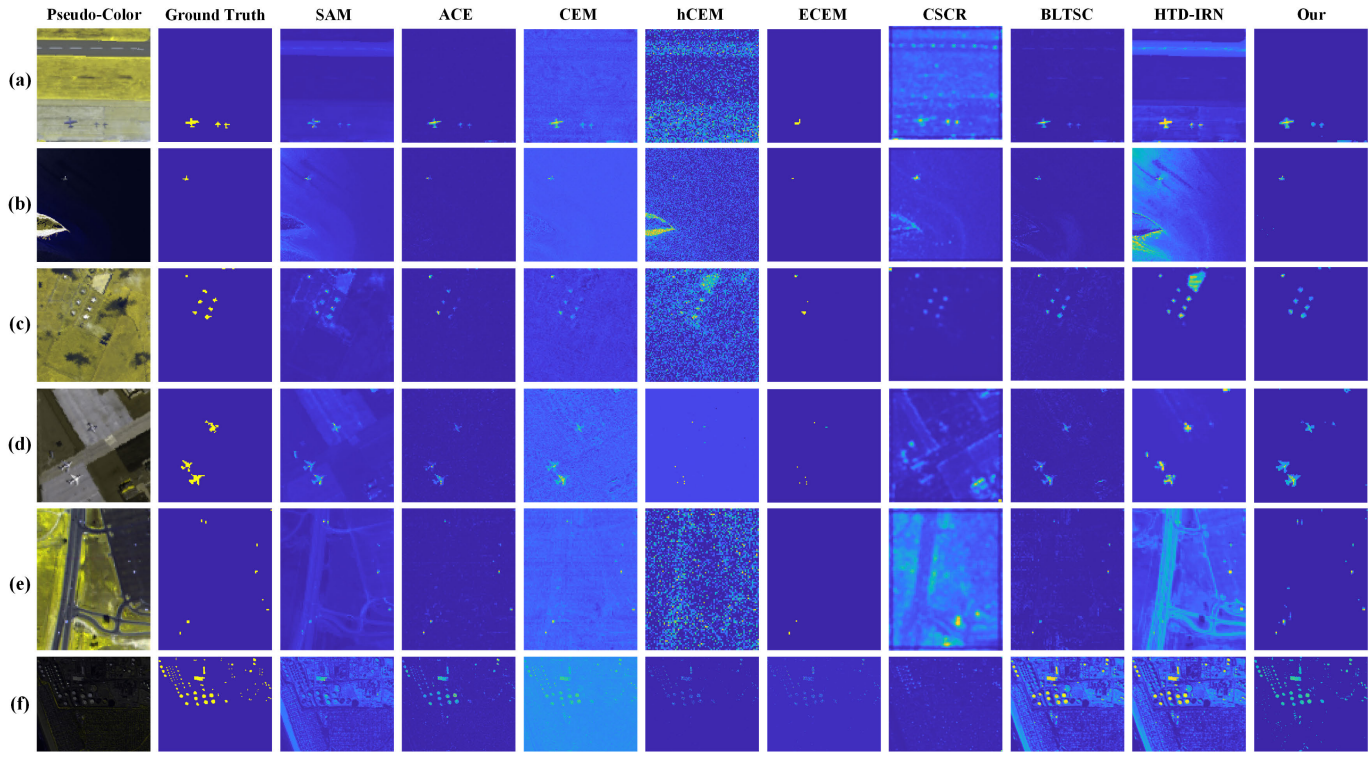


Fig. 5. Visualization of detection results of nine detection methods on six hyperspectral datasets for HTD. (a) Airport. (b) Beach. (c) Urban. (d) San Diego. (e) HYDICE. (f) EI Segundo.

TABLE II
AUC VALUES OF NINE DETECTION METHODS ON SIX HYPERSPECTRAL DATASETS FOR HTD

Methods	Airport		Beach		Urban		San Diego		HYDICE		EI Segundo	
	AUC_1	AUC_2	AUC_1	AUC_2	AUC_1	AUC_2	AUC_1	AUC_2	AUC_1	AUC_2	AUC_1	AUC_2
SAM	0.98268	0.02645	0.98302	0.05615	0.99450	0.02663	0.99518	0.04154	0.99103	0.03740	0.88378	0.08904
ACE	0.95583	0.00237	0.83649	0.00334	0.93894	0.00365	0.93941	0.00578	0.97184	0.00511	0.97399	0.01079
CEM	0.94089	0.14025	0.87960	0.15790	0.96619	0.08563	0.98154	0.17480	0.99062	0.18294	0.98680	0.31139
hCEM	0.76090	0.27216	0.81773	0.11849	0.88516	0.21474	0.98199	0.18385	0.88480	0.29888	0.98857	0.31087
E-CEM	0.79541	0.00544	0.79495	0.00233	0.85960	0.00451	0.76768	0.00573	0.89290	0.00093	0.88279	0.03329
CSCR	0.98637	0.16133	0.99949	0.06048	0.99679	0.00912	0.91555	0.05511	0.53890	0.22134	0.77071	0.01808
BLTSC	0.94177	0.00391	0.99114	0.00518	0.98199	0.00685	0.98728	0.00662	0.99430	0.00815	0.98537	0.11447
HTD-IRN	**0.99138**	0.05967	0.97947	0.16160	0.98937	0.02410	0.98423	0.04598	0.99461	0.18690	0.98079	0.11770
Our	0.99136	**0.00210**	**0.99990**	**0.00016**	**0.99874**	**0.00103**	**0.99803**	**0.00286**	**0.99975**	**0.00083**	**0.99513**	**0.00642**

can capture the shapes of most storage tanks and towers as well as their positions. This is attributed to the wavelet transform in the DWPE module, which can extract the spatial information of the targets clearly. However, other methods are more likely to ignore the targets, get distracted by the background, or lose the shape of the targets. Therefore, the proposed method achieves a satisfactory balance between highlighting the target and suppressing the background.

In order to compare the performance of different detection methods quantitatively, Table II lists the AUC values for all methods, where the highest values are highlighted in bold, and the suboptimal values are underlined. As shown in Table II, the proposed method significantly improves detection accuracy. Except for the Airport dataset, where the AUC_1 value of the proposed method is slightly lower than that of the HTD-IRN method, the proposed method achieves the highest AUC_1 values on other datasets, exceeding 0.99 in precision. For example, on the Beach dataset, the AUC_1 value is 0.99990, which is higher than those of SAM, ACE, CEM, hCEM, E-CEM, CSCR, BLTSC, and HTD-IRN by 0.01688, 0.16341, 0.12030, 0.18217, 0.20495, 0.00041, 0.00876, and 0.02043, respectively. In addition, the AUC_2 values obtained by the proposed method are the lowest, reaching the second decimal place in precision. For instance, on the EI Segundo dataset, the proposed method is the only one that reduces the AUC_2 value to the second decimal place in precision. Note that improvements in the range of 0.01–0.001 are highly significant in HTD. A larger AUC_1 value is deemed to have a better detection performance, and a smaller AUC_2 value indicates that the background is suppressed better in the detection results. Therefore, the proposed method shows a better capability to highlight targets and suppress backgrounds.

Table III shows the experimental results on six HSI datasets, including the coarse results **C** obtained in the CDR module, the prediction results **R** obtained in the inference stage, and the final detection results **Detc** obtained in the

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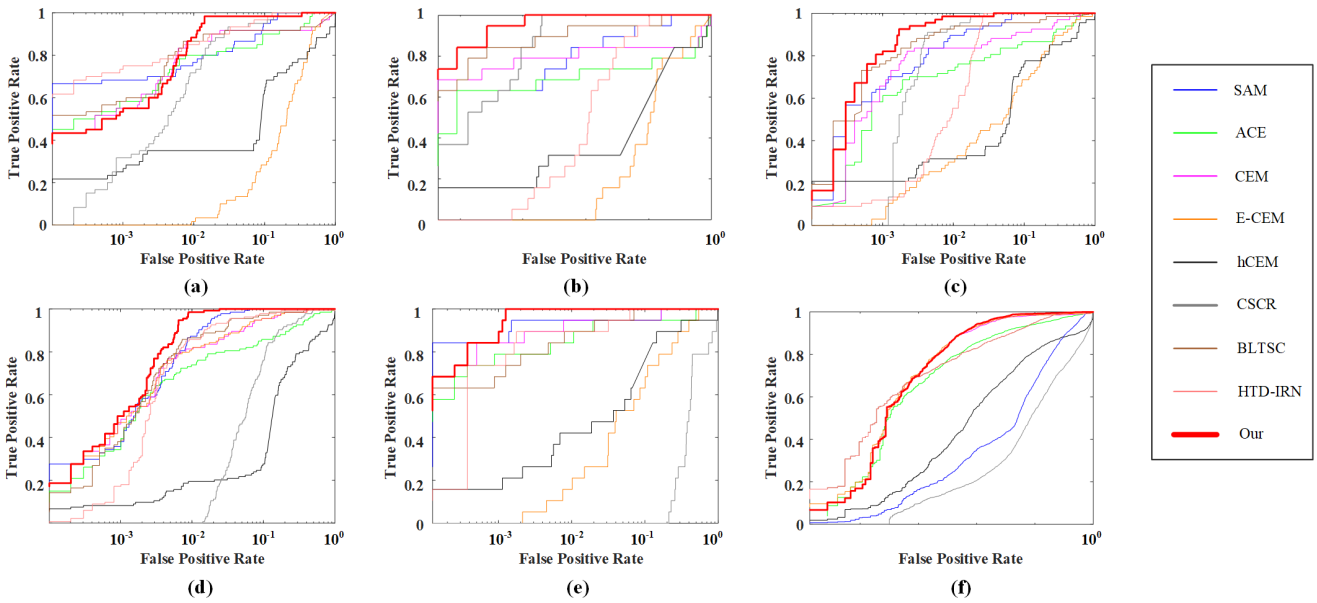


Fig. 6. ROC curves of (FPR and TPR) of detection results from nine detection methods on six hyperspectral datasets for HTD. (a) Airport. (b) Beach. (c) Urban. (d) San Diego. (e) HYDICE. (f) EI Segundo.

TABLE III
EXPERIMENTAL RESULTS OF DIFFERENT STAGES ON SIX HYPERSPECTRAL DATASETS IN THE AUC SCORES

Results	Airport		Beach		Urban		San Diego		HYDICE		EI Segundo	
	AUC_1	AUC_2	AUC_1	AUC_2	AUC_1	AUC_2	AUC_1	AUC_2	AUC_1	AUC_2	AUC_1	AUC_2
C	0.94089	0.14025	0.87960	0.15790	0.96619	0.08563	0.98154	0.17480	0.99062	0.18294	0.98680	0.31139
R	**0.99141**	0.00786	0.99981	0.00067	0.99852	0.00592	0.99772	0.01161	0.99967	0.00316	0.99431	0.01653
Detc	0.99136	**0.00210**	**0.99990**	**0.00016**	**0.99874**	**0.00103**	**0.99803**	**0.00286**	**0.99975**	**0.00083**	**0.99513**	**0.00642**

detection stage. As illustrated in AUC_1, the proposed method achieves higher AUC_1 values in both the inference and detection stages compared with **C**. Meanwhile, as illustrated in AUC_2, in the detection stage, the pixel-level fusion in CDPF significantly strengthens the background suppressibility of the whole framework. In addition, as observed from Table III, the original CEM detector can detect potential targets from spectral information. However, it does not fully exploit the spatial correlation among pixels in HSIs, resulting in lower AUC_1 and AUC_2 values compared with **R** and **Detc**. In the inference stage, our method can extract spectral–spatial features using the DWPE module and the pretrained encoder, which realizes better target–background separation. Therefore, the AUC_1 and AUC_2 values of **R** outperform those of **C**. In the detection stage, the proposed method applies the prediction results to the coarse detection results via the CDPF module. This procedure effectively suppresses the background by combining the discriminative feature from the spectral and spatial domains. Although on the Airport dataset, the AUC_1 value of **Detc** is slightly lower than that of **R**, the AUC_2 value still shows a significant advantage. Therefore, it can be concluded that our method makes full use of spectral–spatial information and produces detection results with complementary information to coarse results.

Fig. 6 presents the ROC curves of (FPR, TPR) of detection results from different methods on six datasets. Note that the closer to the top-left corner of these curves, the better the detection performance is. It can be observed that the proposed method always remains over those compared methods, except for the Airport dataset. On the Airport dataset, the proposed method initially shows a lower detection probability than other competing methods. However, when FPR exceeds the value of 0.01, the red curve representing the proposed method obviously surpasses other curves. In general, the proposed method proves to be more reliable than others.

For better analyzing the separability of the target and background among different methods, the target–background separability plot is displayed in Fig. 7. The red and green boxes represent the distribution of detection values for the targets and the background, respectively. Also, the positions of red and green boxes represent the separability of targets and backgrounds. The greater the gap between the red box and the green box, the better the target–background separability of the detector. It is evident that the proposed method outperforms other methods in separating targets from background, particularly on the EI Segundo dataset, where the distance between two boxes is larger.

D. Ablation Experiment

To explore the effectiveness of these key components in the proposed method, ablation analyses are performed and analyzed by removing each component from the proposed method. There are three key components in the ablation analyses: the DWPE module, the SC-CE loss, and the CDPF

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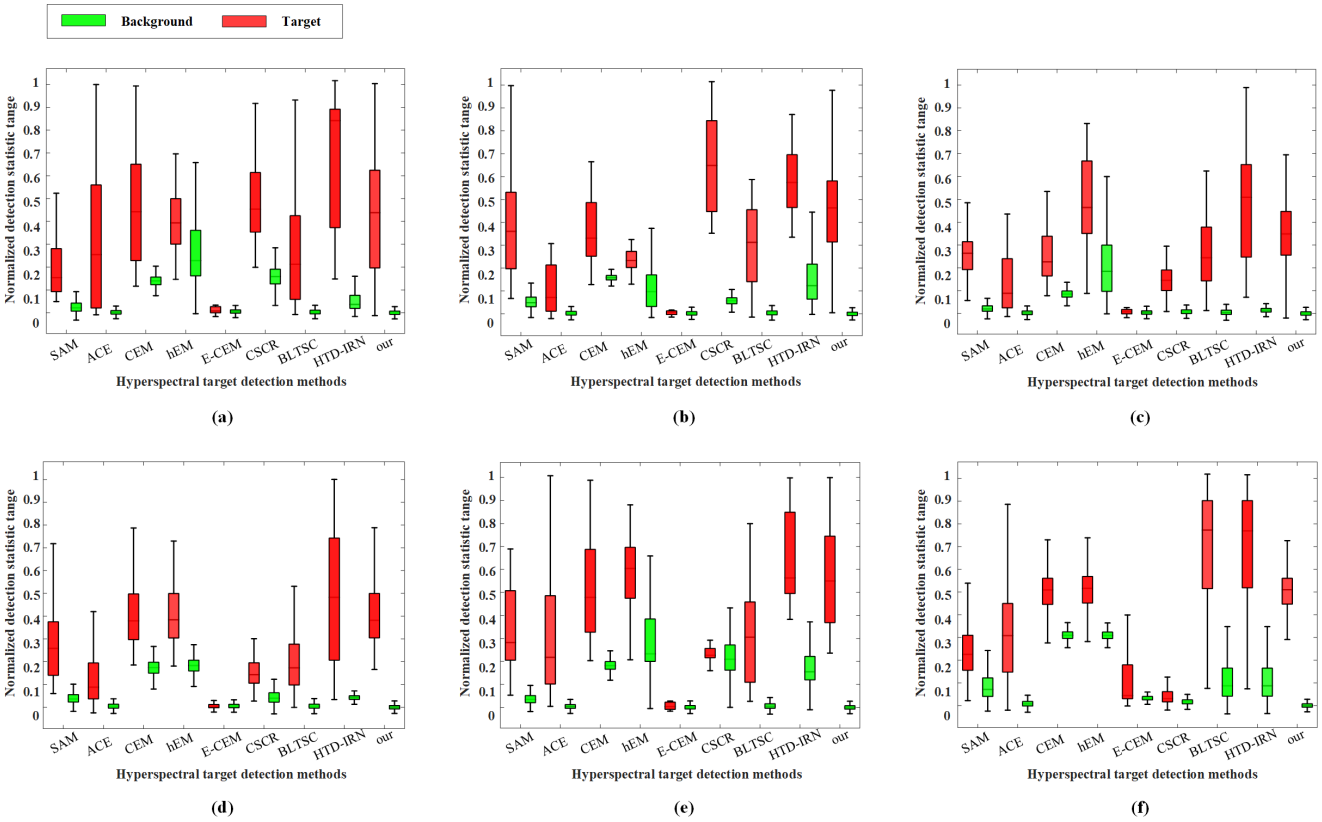


Fig. 7. Separability maps of detection results from nine detection methods on six hyperspectral datasets for HTD. (a) Airport. (b) Beach. (c) Urban. (d) San Diego. (e) HYDICE. (f) EI Segundo.

TABLE IV
AUC SCORES OF THE PROPOSED COMPONENTS ON FOUR DATASETS

Components			Beach		Urban		San Diego		HYDICE	
DWPE	SC-CE	CDPF	AUC_1	AUC_2	AUC_1	AUC_2	AUC_1	AUC_2	AUC_1	AUC_2
✓			0.99985	0.00083	0.99797	0.00529	0.99707	0.00966	0.99945	0.00291
✓		✓	0.99987	0.00020	0.99823	0.00111	0.99731	0.00311	0.99970	0.00088
	✓		0.99927	0.00172	0.99770	0.00909	0.99369	0.01147	0.99307	0.01133
	✓	✓	0.99933	0.00037	0.99799	0.00151	0.99477	0.00302	0.99585	0.00256
✓	✓		0.99981	0.00067	0.99852	0.00592	0.99772	0.01161	0.99967	0.00316
✓	✓	✓	**0.99990**	**0.00016**	**0.99874**	**0.00103**	**0.99803**	**0.00286**	**0.99975**	**0.00083**

module. The detection performance of different components of the proposed method is evaluated on four real HSI datasets: Beach, Urban, San Diego, and HYDICE. To mitigate the effects of random factors, five consecutive executions of code were performed, and the average AUC values were obtained. Table IV shows the AUC scores of the ablation study of the proposed method, where the highest values are highlighted in bold.

As shown in Table IV, it can be found that each component of the proposed framework can effectively improve the detection performance. Specifically, for all datasets, using only the DWPE module presents better detection accuracy than only using the SC-CE loss. This can be attributed to the fact that on the one hand, DWPE can transform the feature patch into the fused feature, which contains rich spectral–spatial information for detection. On the other hand, the SC-CE loss is only responsible for optimizing the loss trend during training, which brings less positive effects on detection performance than the DWPE module. Then, when the CDPF module is added to the DWPE module and the SC-CE loss, the detection performance is further prompted, which shows that the pixel-level fusion can refine the original detection result effectively. Finally, when all the key components are applied, the detection results on four datasets achieve the best effects, which manifest the effectiveness of each component of the proposed method.

E. Parameter Sensitivity

This section analyzes three crucial parameters of the proposed method, including the number of principal components L in PCA and the feature patch size P in the preprocessing, and the mask ratio Q during the pretraining stage. Also, the model complexity of the DL-based method is also analyzed. Comprehensive experiments were performed on four real HSI

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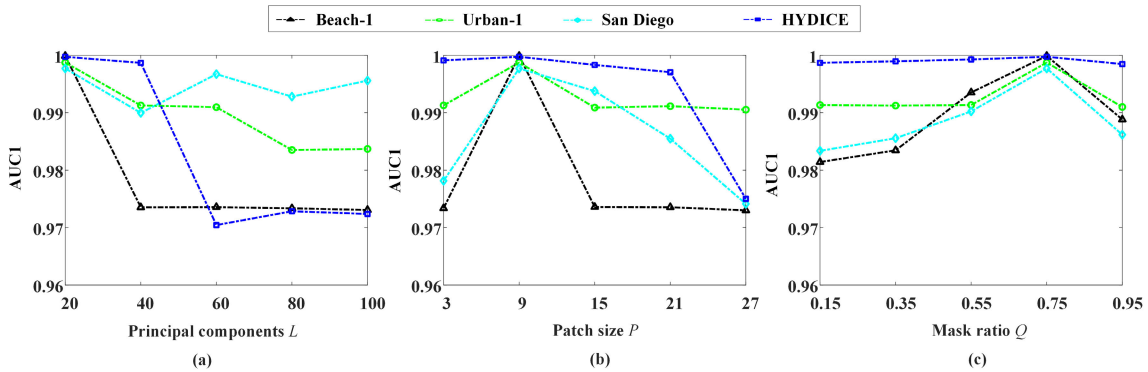


Fig. 8. Parameter sensitivity analysis on four hyperspectral datasets for HTD. (a) Principal components L. (b) Patch size P. (c) Mask ratio Q.

datasets: Beach, Urban, San Diego, and HYDICE. To mitigate the effects of random factors, five consecutive executions of code were performed, and the average AUC values were obtained.

1) Influence of Principal Components: PCA is used to reduce the spectral redundancy of the input HSI. The number of principal components L in PCA is ranging from 20 to 100. Fig. 8(a) shows how L affects detection performance on different datasets. As the number of principal components increases, the AUC_1 value decreases on the Beach, Urban, and HYDICE datasets. However, on the San Diego dataset, the AUC_1 value shows slight fluctuations but remains lower than the AUC_1 value when L is 20. In addition, using a larger number of principal components increases both time consumption and the risk of memory overflow. It can be found that the best detection performance on all four datasets is achieved when L is 20. Therefore, we set the number of principal components as 20 for the experiments in this work.

2) Influence of Patch Size: The feature patch size P has a significant influence on the local receptive field during the model training. The patch size is set to 3, 9, 15, 21, and 27. Fig. 8(b) shows how P affects detection performance on different datasets. It can be observed that when P is set to 9, the value of AUC_1 reaches its peak. A smaller patch size limits the ability of the model to capture neighboring information during training, while a larger patch size prevents the model from extracting target features effectively. Therefore, we set the size of feature patches as 9 for the experiments in this work.

3) Influence of Mask Ratio: The mask ratio Q determines the number of visible spectral–spatial vectors in the fused feature $\mathbf{x}_{ss}$ with $\mathbf{E}_{pos}$, which affects the FE ability of encoder. The mask ratio Q is varied from 0.15 to 0.90 in increments of 0.15. Fig. 8(c) shows how Q affects detection performance on different datasets. As shown in Fig. 8(c), the AUC_1 value increases with Q, reaching a maximum when Q is 0.75, and then decreases as Q continues to rise. A smaller mask ratio facilitates the reconstruction by containing more spectral–spatial vectors in the fused feature $\mathbf{x}_{ss}$ with $\mathbf{E}_{pos}$, which results in the weak FE ability of encoder, resulting in lower AUC_1 values. In addition, when the mask ratio exceeds 0.75, the AUC_1 value starts to decrease as fewer spectral–spatial vectors in the fused feature $\mathbf{x}_{ss}$ with $\mathbf{E}_{pos}$ are contained. Therefore, we set the mask ratio as 0.75 for the experiments in this work.

TABLE V
COMPARISON OF DL-BASED METHODS ON COMPLEXITY

Dataset	Complexity	BLTSC	HTD-IRN	Our
Beach	Parameters (M)	1.57	0.41	**0.18**
	FLOPs (M)	**2.837**	2355.12	4975.12
Urban	Parameters (M)	1.57	0.41	**0.18**
	FLOPs (M)	**2.62**	2185.64	4867.25
San Diego	Parameters (M)	1.57	0.41	**0.18**
	FLOPs (M)	**2.61**	2176.46	4856.99
HYDICE	Parameters (M)	1.57	0.41	**0.18**
	FLOPs (M)	**2.59**	1902.15	4767.58

4) Model Complexity Analysis: The model complexity mainly includes space complexity and time complexity. Therefore, we utilize the number of model parameters and the floating-point operations (FLOPs) to measure space and time complexity, respectively. Table V lists the quantitative comparison results of the model complexity of all DL-based methods on four HSI datasets. Note that we only calculate the model complexity in the testing stage. It can be observed that BLTSC typically has fewer parameters and FLOPs due to it is a simple model based on adversarial AE (AAE). In addition, HTD-IRN transforms the HTD subspace representation model (SRM) into a subspace representation network (SRN) and enhances nonlinear features with a multiscale transformer, resulting in a deeper and more complex network structure. Therefore, HTD-IRN has more parameters and FLOPs than BLTSC. In contrast, our method maintains the simplest network structure only using a transformer to extract HSI features. As a result, our method has the fewest parameters, but due to the more complex GMSA, it has the highest FLOPs among all methods.

IV. CONCLUSION

This article proposes an effective HTD method based on GSSL to reduce the reliance on large-scale datasets for pretraining and explore the integration of WT and CNNs for HTD. In the preprocessing, the proposed DT module and CDR module make full use of the input HSI to produce feature patches as training samples for subsequent pretraining and fine-tuning. In the pretraining stage based on GSSL, the proposed DWPE module transforms each input

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feature patch into the fused feature, which enhances the spectral–spatial representation of the input HSI. In the fine-tuning and detection stage, the proposed SC-CE loss improves the detection accuracy by mitigating background interference during fine-tuning. And for detection, the proposed CDPF module suppresses the interference from complex and variable backgrounds. Experimental results on six real HSI datasets demonstrate the superiority of the proposed method. Future work could explore integrating wavelet transform into the attention mechanism to further enhance the integration of spectral and spatial features.

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